



Faculty of Computer Applications

**MCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code: 05MC0108
(DBMS)**

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1	Write a PL/SQL block that find 1. Area Of Circle 2. Area Of Rectangle 3. Circumference of Circle
2	Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).
3	Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).
4	1. Write a program that prints value 1 to 100 numbers using FOR LOOP. 2. Write a program that prints value 1 to 100 number using LOOP Command. 3. Write a program that prints value 1 to 100 number using WHILE LOOP Command
5	Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno, Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.
6	Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.
7	Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.
8	Write a PL/SQL block to delete the record of employee for given EID.
9	Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.
10	Write a PL/SQL block to calculate the square and cube of the given number.

1. Write a PL/SQL block that find:

1. Area of Circle
 2. Area of Rectangle
 3. Circumference of Circle
-

/*

Write a PL/SQL block that find

- 1.Area of circle(π *radius*radius)
- 2.Area of Rectangle(length*width)
- 3.Circumference of circle($2*\pi*RADIUS$)

*/

ACCEPT CHOICE PROMPT "Enter Your Choice: 1.Area of circle ,2.Area of Rectangle
,3.Circumference of circle "

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

DECLARE

CHOICE NUMBER:=&CHOICE;

PI NUMBER(7,2):=3.14;

RADIUS NUMBER(5);

LENGTH NUMBER(5);

WIDTH NUMBER(5);

BEGIN

--CHOICE:=&CHOICE;

DBMS_OUTPUT.PUT_LINE(' ');

IF (CHOICE=1) THEN

RADIUS:=&RADIUS;

DBMS_OUTPUT.PUT_LINE('Area of circle is : ' || π *(POWER(RADIUS,2)));

ELSIF (CHOICE=2) THEN

LENGTH:=&LENGTH;

WIDTH:=&WIDTH;

DBMS_OUTPUT.PUT_LINE('Area of Rectangle is : ' || LENGTH*WIDTH);

ELSIF (CHOICE=3) THEN

DBMS_OUTPUT.PUT_LINE('Circumference of circle is : ' || $2*\pi*RADIUS$);

ELSE

```
-
        DBMS_OUTPUT.PUT_LINE('Invalid Choice');
    END IF;
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 1
Enter value for radius: 12
Enter value for length: 8
Enter value for width: 5
Area of circle is : 452.16
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 2
Enter value for radius: 12
Enter value for length: 45
Enter value for width: 40
Area of Rectangle is : 1800
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p1.sql
Enter Your Choice: 1.Area of circle ,2.Area of Rectangle ,3.Circumference of circle :- 3
Enter value for radius: 10
Enter value for length: 40
Enter value for width: 45
Circumference of circle is : 62.8
```

2. Write a PL/SQL block to accept product name, qty and price from user and then calculate discount in Rs. based on the given (%).

```
/*
Write a PL/SQL block to accept product name ,qty and price from user
and then calculate discount in Rs.based on the given (%)
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF

ACCEPT PN PROMPT "Enter product name :- "
DECLARE
    P_NAME      VARCHAR(20);
    QTY   NUMBER(5);
    PRICE  NUMBER(7,2);
    TP     NUMBER(7,2);
    DISCOUNT  NUMBER(5);
    FP     NUMBER(7,2);
BEGIN
    P_NAME:='&PN';
    QTY:=&QTY;
    PRICE:=&PRICE;
    DISCOUNT:=&DISCOUNT;

    TP:=PRICE*QTY;
    FP:=TP-(TP*DISCOUNT/100);
    DBMS_OUTPUT.PUT_LINE(' ');
    DBMS_OUTPUT.PUT_LINE('Product name is := ' || P_NAME);
    DBMS_OUTPUT.PUT_LINE('Total qty is := ' || QTY);
    DBMS_OUTPUT.PUT_LINE('Price is := ' || PRICE);
    DBMS_OUTPUT.PUT_LINE('Total Price is := ' || TP);
    DBMS_OUTPUT.PUT_LINE('Discount is := ' || DISCOUNT);
    DBMS_OUTPUT.PUT_LINE('Final Price is := ' || FP);

END;
/
```

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p2.sql
Enter product name :- key-board
Enter value for qty: 3
Enter value for price: 200
Enter value for discount: 12

Product name is := key-board
Total qty is := 3
Price is := 200
Total Price is := 600
Discount is := 12
Final Price is := 528
```

3. Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).

/*

Write a PL/SQL block to calculate the total, percentage and grade of student based on his/her Rollno from RESULT table. (Create RESULT table with Rollno, Name, Sub1, Sub2, Sub3, Sub4, Sub5, Total, Per, Grade attributes with appropriate data type).

*/

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

DECLARE

ROLL_NO NUMBER(5);

NAME VARCHAR2(30);

SUB1 NUMBER(3);

SUB2 NUMBER(3);

SUB3 NUMBER(3);

SUB4 NUMBER(3);

SUB5 NUMBER(3);

TOTAL NUMBER(5);

PER NUMBER(7,2);

GRADE VARCHAR2(30);

BEGIN

ROLL_NO:=&ROLL_NO;

NAME:='&NAME';

SUB1:=&SUB1;

SUB2 :=&SUB2 ;

SUB3 :=&SUB3 ;

SUB4:=&SUB4;

SUB5:=&SUB5;

TOTAL:=SUB1+SUB2+SUB3+SUB4+SUB5;

PER:=TOTAL/5;

IF (PER>=75) THEN

GRADE:='O';

ELSIF (PER<=75)AND(PER>=70) THEN

```
        GRADE:='A';
    ELSIF (PER<=60)AND(PER>=70) THEN
        GRADE:='B';
    ELSIF (PER<=50)AND(PER>=60) THEN
        GRADE:='C';
    ELSE
        GRADE:='PASS CLASS';
    END IF;

    INSERT INTO
    RESULT(ROLLNO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE)VALUES(RO
    LL_NO,NAME,SUB1,SUB2,SUB3,SUB4,SUB5,TOTAL,PER,GRADE);

    DBMS_OUTPUT.PUT_LINE(' ');
    DBMS_OUTPUT.PUT_LINE('Roll no:-' || ROLL_NO);
    DBMS_OUTPUT.PUT_LINE('Name:-' || NAME);
    DBMS_OUTPUT.PUT_LINE(' ');
    DBMS_OUTPUT.PUT_LINE('-----Result-----');
    DBMS_OUTPUT.PUT_LINE('Marks of sub1:-' || SUB1);
    DBMS_OUTPUT.PUT_LINE('Marks of sub2:-' || SUB2);
    DBMS_OUTPUT.PUT_LINE('Marks of sub3:-' || SUB3);
    DBMS_OUTPUT.PUT_LINE('Marks of sub4:-' || SUB4);
    DBMS_OUTPUT.PUT_LINE('Marks of sub5:-' || SUB5);

    DBMS_OUTPUT.PUT_LINE(' ');
    DBMS_OUTPUT.PUT_LINE('-----');
    DBMS_OUTPUT.PUT_LINE('Your total marks : ' || TOTAL);
    DBMS_OUTPUT.PUT_LINE('Percentage : ' || PER || '%');
    DBMS_OUTPUT.PUT_LINE('Grade:' || GRADE);
END;
/

SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON
```


Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p3.sql
Enter value for roll_no: 17
Enter value for name: kunjal
Enter value for sub1: 80
Enter value for sub2: 90
Enter value for sub3: 88
Enter value for sub4: 96
Enter value for sub5: 95
```

```
Roll no:-17
Name:-kunjal
```

```
-----Result-----
Marks of sub1:-80
Marks of sub2:-90
Marks of sub3:-88
Marks of sub4:-96
Marks of sub5:-95
```

```
-----
Your total marks : 449
Percentage :89.8%
Grade:0
```



```

DBMS_OUTPUT.PUT_LINE('');
DBMS_OUTPUT.PUT_LINE('--- WHILE LOOP ---');
i := 1;
WHILE i <= 100 LOOP
    DBMS_OUTPUT.PUT(i || ' ');
    i := i + 1;
END LOOP;
DBMS_OUTPUT.PUT_LINE('');
END;
/
SET SERVEROUTPUT OFF
SET VERIFY ON
SET FEEDBACK ON

```

Output:

```

SQL> @ D:\Kunjal\DBMS\Unit_1\p4.sql
--- FOR LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- SIMPLE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100
--- WHILE LOOP ---
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30
31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57
58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84
85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

```

5. Write a PL/SQL block which displays gross salary of employees as per user input EID. (Consider EMP table with EID, EName, Deptno, Deptname Gender, Age, BasicSal) with appropriate data types.) Gross_Salary: BASICSAL + (DA + HRA + Medical) – PF. Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.

/*

Write a PL/SQL block which displays gross salary of employees as per user input EID.

Consider EMP table with EID, EName, Deptno, Deptname, Gender, Age, BasicSal with appropriate data types.

Gross_Salary:= BASICSAL + (DA + HRA + Medical) – PF.

Rules: HRA = 15% of basic, DA = 50% of basic, Medical = Rs. 500, PF = 10% of basic.

Here for practical we have used Scott users inbuilt EMP table.

*/

SET SERVEROUTPUT ON

SET FEEDBACK OFF

SET VERIFY OFF

DECLARE

V_EID EMP.EID%TYPE;

V_NAME EMP.ENAME%TYPE;

BASIC_SAL EMP.BASICSAL%TYPE;

DA NUMBER;

HRA NUMBER;

MEDICAL NUMBER := 500;

PF NUMBER;

GROSS_SAL NUMBER;

BEGIN

V_EID := &EID;

SELECT ENAME, BASICSAL

INTO V_NAME, BASIC_SAL

FROM EMP

WHERE EID = V_EID;

DA := BASIC_SAL * 0.50;

```

HRA := BASIC_SAL * 0.15;
PF := BASIC_SAL * 0.10;
GROSS_SAL := BASIC_SAL + DA + HRA + MEDICAL - PF;

DBMS_OUTPUT.PUT_LINE('----- SALARY SLIP -----');
DBMS_OUTPUT.PUT_LINE('Employee ID   : ' || V_EID);
DBMS_OUTPUT.PUT_LINE('Employee Name  : ' || V_NAME);
DBMS_OUTPUT.PUT_LINE('Basic Salary   : Rs. ' || BASIC_SAL);
DBMS_OUTPUT.PUT_LINE('DA Amount      : Rs. ' || DA);
DBMS_OUTPUT.PUT_LINE('HRA Amount     : Rs. ' || HRA);
DBMS_OUTPUT.PUT_LINE('Medical Amount : Rs. ' || MEDICAL);
DBMS_OUTPUT.PUT_LINE('Provident Fund : Rs. ' || PF);
DBMS_OUTPUT.PUT_LINE('-----');
DBMS_OUTPUT.PUT_LINE('Gross Salary   : Rs. ' || GROSS_SAL);
DBMS_OUTPUT.PUT_LINE('-----');
EXCEPTION
  WHEN NO_DATA_FOUND THEN
    DBMS_OUTPUT.PUT_LINE(
      'Employee record for given EID - ' || V_EID || ' does not exist'
    );
END;
/

SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON

```

Output:

```

SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
----- SALARY SLIP -----
Employee ID      : 105
Employee Name    : PRIYA
Basic Salary     : Rs. 32000
DA Amount        : Rs. 16000
HRA Amount       : Rs. 4800
Medical Amount   : Rs. 500
Provident Fund    : Rs. 3200
-----
Gross Salary     : Rs. 50100
-----

```

6. Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

/*

Write a PL/SQL block which accepts measurement in feet and displays it in cm, inch and meter.

Formulas:-

1)Inches :-feet * 12

2)Centimeters :- feet * 30.48

3)Meters:-feet*0.3045

*/

SET SERVEROUTPUT ON

SET VERIFY OFF

SET FEEDBACK OFF

DECLARE

FEET NUMBER;

CM NUMBER;

INCH NUMBER;

METER NUMBER;

BEGIN

FEET := &FEET;

CM := FEET * 30.48;

INCH := FEET * 12;

METER := FEET * 0.3048;

DBMS_OUTPUT.PUT_LINE('Feet : ' || FEET);

DBMS_OUTPUT.PUT_LINE('Centimeter : ' || CM);

DBMS_OUTPUT.PUT_LINE('Inch : ' || INCH);

DBMS_OUTPUT.PUT_LINE('Meter : ' || METER);

END;

/

SET SERVEROUTPUT OFF

SET VERIFY ON

SET FEEDBACK ON

-

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p5.sql
Enter value for eid: 105
----- SALARY SLIP -----
Employee ID      : 105
Employee Name    : PRIYA
Basic Salary     : Rs. 32000
DA Amount        : Rs. 16000
HRA Amount       : Rs. 4800
Medical Amount   : Rs. 500
Provident Fund    : Rs. 3200
-----
Gross Salary     : Rs. 50100
-----
```

7. Write a PL/SQL block which converts temperature from Celsius to Fahrenheit.

```
/*
Write a PL/SQL block which converts temperature from Celsius to
Fahrenheit.
Formulas:-
    -Celsius to Fahrenheit :-(Celsius *9/2)+32
    -Fahrenheit to Celsius :-(Fahrenheit-32)/5/9
*/

SET SERVEROUTPUT ON
SET VERIFY OFF
SET FEEDBACK OFF
ACCEPT CH PROMPT 'Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to
Celsius: '
DECLARE
    CELSIUS NUMBER(10,2);
    FAHRENHEIT NUMBER(10,2);
    CHOICE NUMBER(10);
BEGIN
    CHOICE := &CH;
    IF CHOICE = 1 THEN
        CELSIUS := &CELSIUS;
        FAHRENHEIT := (CELSIUS * 9 / 5) + 32;

        DBMS_OUTPUT.PUT_LINE('Value of FAHRENHEIT :- ' || FAHRENHEIT);

    ELSIF CHOICE = 2 THEN
        FAHRENHEIT := &FAHRENHEIT;
        CELSIUS := (FAHRENHEIT - 32) * 5 / 9;
        DBMS_OUTPUT.PUT_LINE('Value of CELSIUS :- ' || CELSIUS || 'C');
    ELSE
        DBMS_OUTPUT.PUT_LINE('INVALID CHOICE... Enter 1 or 2');
    END IF;
END;
/
```


Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 1
Enter value for celsius: 12
Enter value for fahrenheit: 30
Value of FAHRENHEIT :- 53.6
```

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p7.sql
Enter your choice :- 1. Celsius to Fahrenheit, 2. Fahrenheit to Celsius: 2
Enter value for celsius: -45
Enter value for fahrenheit: 45
Value of CELSIUS :- 7.22C
SQL>
```

8. Write a PL/SQL block to delete the record of employee for given EID.

```
/*
Write a PL/SQL block to remove an employee record
for the given Employee ID.
*/

SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
    EMP_ID EMP.EID%TYPE;
    CNT NUMBER;
BEGIN
    EMP_ID := &EID;

    SELECT COUNT(*) INTO CNT FROM EMP WHERE EID = EMP_ID;

    IF CNT > 0 THEN
        DELETE FROM EMP WHERE EID = EMP_ID;

        COMMIT;
        DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' deleted successfully.'
);
    ELSE
        DBMS_OUTPUT.PUT_LINE('Employee ID ' || EMP_ID || ' does not exist.' );
    END IF;

END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
value of EID: 105
SQL> @ D:\Kunjai\DBMS\Unit_1\p8.sql
Enter value for eid: 105
Employee ID 105 deleted successfully.
SQL>
```

9. Write a PL/SQL block that calculates the simple interest based on given principal amount, rate of interest and number of years.

```
/* Write a PL/SQL block to calculate Simple Interest and Total Amount. */
```

```
SET SERVEROUTPUT ON
SET FEEDBACK OFF
SET VERIFY OFF
DECLARE
    P NUMBER;
    R NUMBER;
    T NUMBER;
    SI NUMBER;
    AMOUNT NUMBER;
BEGIN
    P := &P;
    R := &R;
    T := &T;
    SI := (P * R * T) / 100;
    AMOUNT := P + SI;
    DBMS_OUTPUT.PUT_LINE('=====');
    DBMS_OUTPUT.PUT_LINE('Principal Amount : ' || P);
    DBMS_OUTPUT.PUT_LINE('Rate of Interest : ' || R || '%');
    DBMS_OUTPUT.PUT_LINE('Number of Years : ' || T);
    DBMS_OUTPUT.PUT_LINE('Simple Interest : ' || SI);
    DBMS_OUTPUT.PUT_LINE('Total Amount : ' || AMOUNT);
END;
/
SET SERVEROUTPUT OFF
SET FEEDBACK ON
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjal\DBMS\Unit_1\p9.sql
Enter value for p: 12000
Enter value for r: 10
Enter value for t: 5
=====
Principal Amount : 12000
Rate of Interest : 10%
Number of Years : 5
Simple Interest : 6000
Total Amount : 18000
```

10. Write a PL/SQL block to calculate the square and cube of the given number.

```
/*  
Write a PL/SQL block to calculate the square and cube  
of the given number.  
*/
```

```
SET SERVEROUTPUT ON  
SET FEEDBACK OFF  
SET VERIFY OFF
```

```
DECLARE  
  N NUMBER;  
  SQUARE NUMBER;  
  CUBE NUMBER;  
BEGIN  
  N := &N;  
  SQUARE := N * N;  
  CUBE := N * N * N;  
  
  DBMS_OUTPUT.PUT_LINE('Given Number : ' || N);  
  DBMS_OUTPUT.PUT_LINE('Square      : ' || SQUARE);  
  DBMS_OUTPUT.PUT_LINE('Cube       : ' || CUBE);  
  
END;  
/
```

```
SET SERVEROUTPUT OFF  
SET FEEDBACK ON  
SET VERIFY ON
```

Output:

```
SQL> @ D:\Kunjai\DBMS\Unit_1\p10.sql  
Enter value for n: 20  
Given Number : 20  
Square       : 400  
Cube        : 8000  
SQL> _
```